

PROJECT NAME: MINDSYNC

PROJECT TITLE

Mental Health Assessment and Support System

PROBLEM STATEMENT

Problems like stress, anxiety, and depression have become very prevalent in today's society. Individuals find it difficult to recognize such mental ailments at the initial stages owing to lack of proper knowledge and absence of proper screening tools. This tool tries to develop such an easy and user-friendly tool which would help screen mental health disorders among the people.

WHY DOES THIS PROBLEM EXIST?

- Lack of mental health awareness.
- Shortage of mental health professionals.
- Delayed diagnosis and treatment.
- Social stigma prevents people from seeking help.
- Manual screening is time-consuming.

WHO ARE THE STAKEHOLDERS?

Internal Users

- System Administrator
- Mental Health Counselor
- Healthcare Professionals

External Users

- Students
- Employees
- General Public

Expected System Users

- Individuals seeking mental health assessment.
- Counselors and psychologists.

- Healthcare organizations.
- Educational institutions or any Corporate houses also.

OBJECTIVES OF THE PROJECT

1. Provide early detection of mental health disorders.
2. Reduce manual screening efforts.
3. Predict mental health conditions using machine learning algorithms.
4. Improve accessibility to preliminary mental health assessment.
5. Generate prediction reports for users.
6. Support counselors in decision-making.
7. Maintain secure and confidential user information.
8. Increase awareness regarding mental health issues.

EXPECTED OUTPUTS

- User registration and login.
- Mental health questionnaire interface.
- Predicted mental health disorder category.
- Accuracy and confidence score.
- Downloadable assessment report.
- Admin dashboard for monitoring.
- User history and prediction records.

HANDS-ON ACTIVITY

STRATEGIC INFORMATION

- Long-term mental health trends.
- Disorder prevalence analysis.
- Mental health patterns based on age, gender, occupation, and other demographic details.

TACTICAL INFORMATION

- Monthly prediction statistics.
- User engagement reports.
- Counselor workload planning.
- Assessment completion reports.
- User activity monitoring.

- Feedback analysis reports.
- Questionnaire response summaries.

OPERATIONAL INFORMATION

- Daily questionnaire submissions.
- Individual prediction results.
- User account management.
- Report generation and downloads.
- Daily user registration records.
- System administrator monitoring activities.
- Age-wise user assessment records.
- Daily login activity records.

FEASIBILITY ANALYSIS

Technical Feasibility

This project is technically feasible as it uses various web technologies, databases, and software packages. The system can perform questionnaire-based assessment and age-based analysis of answers provided by users.

Economic Feasibility

The cost of developing and maintaining this project is low. It will bring advantages in form of providing a simple and accessible way for conducting a mental health assessment.

Expected Benefits:

- Quick mental health assessment
- Reduced manual effort
- Better user awareness

Operational Feasibility

This system will be easy to use for the target users, such as students, employees, and regular people. Users will be able to fill out questionnaires and get their assessments performed.

Schedule Feasibility

This project can be done in the available semester period due to the availability of necessary resources and technologies.

SRS Worksheet

Purpose:

To serve as an easy-to-use tool for evaluation of mental well-being through questionnaire-based evaluations and report generation on the results of the evaluations.

Scope:

The system will collect user data, perform the evaluations, generate results, and keep the records.

User Profiles:

- Basic computer or mobile device skills.
- People of all age groups.
- People who want to do self-evaluation and well-being monitoring.
- System administrators.

Functional Requirements

- 1: User must register and login into the system.
- 2: User must answer the mental health questionnaire.
- 3: User must choose frequency of assessments.
- 4: User must view assessment results and be notified about report availability.

Non-Functional Requirements

- 1: User Support - The system shall have good instruction and user support.
- 2: Reliability – The system shall be operational without hanging or crashing while doing the evaluation.

3: Navigation – The system shall have smooth and easy navigation from one page to another.

ETHICAL MEASURES

User Ethics:

Users should give truthful responses to questionnaires and choose proper assessment intervals.

Business Ethics:

Companies should utilize assessment results ethically and only for mental well-being purposes.

Regulatory Ethics (Domain Specific):

The system should adhere to the mental wellness guidelines that apply in India.

Data and Privacy:

Information about users and the assessments done should remain private.

AI and Technology Related:

The system should allow both individual and group assessment. Schools and companies can utilize group assessments to determine their general mental wellness status without breaching user confidentiality. The system is meant for assessment and awareness purposes only, not for making medical diagnoses and prescribing medicines.

Development Ethics:

Developers should consider system reliability and user safety when developing and maintaining the system.

USE CASE ACTIVITY

1. CUSTOMER USE CASE:

USE CASE	ACTOR	OBJECTIVE
Create a New Account	Customer	Register a new account in the system.
Login	Customer	Access the system securely.
Update Profile	Customer	Update personal profile information.
View Assessment Questionnaire	Customer	Access the mental health assessment questionnaire.
Complete and Submit Assessment	Customer	Complete and submit the assessment questionnaire.
View Assessment Score and Result	Customer	View the assessment score, wellness level, and recommendations generated by the system.
Request Customized Questionnaire	Customer	Request a customized questionnaire if required.
Receive Customized Questionnaire	Customer	Receive the customized questionnaire from the psychologist.
Complete and Submit Customized Assessment	Customer	Complete and submit the customized assessment questionnaire.
View Updated Assessment Score and Result	Customer	View the updated assessment score, wellness level, and recommendations.
Download Assessment Report	Customer	Download the assessment report for future reference.
Logout	Customer	Exit the system securely.

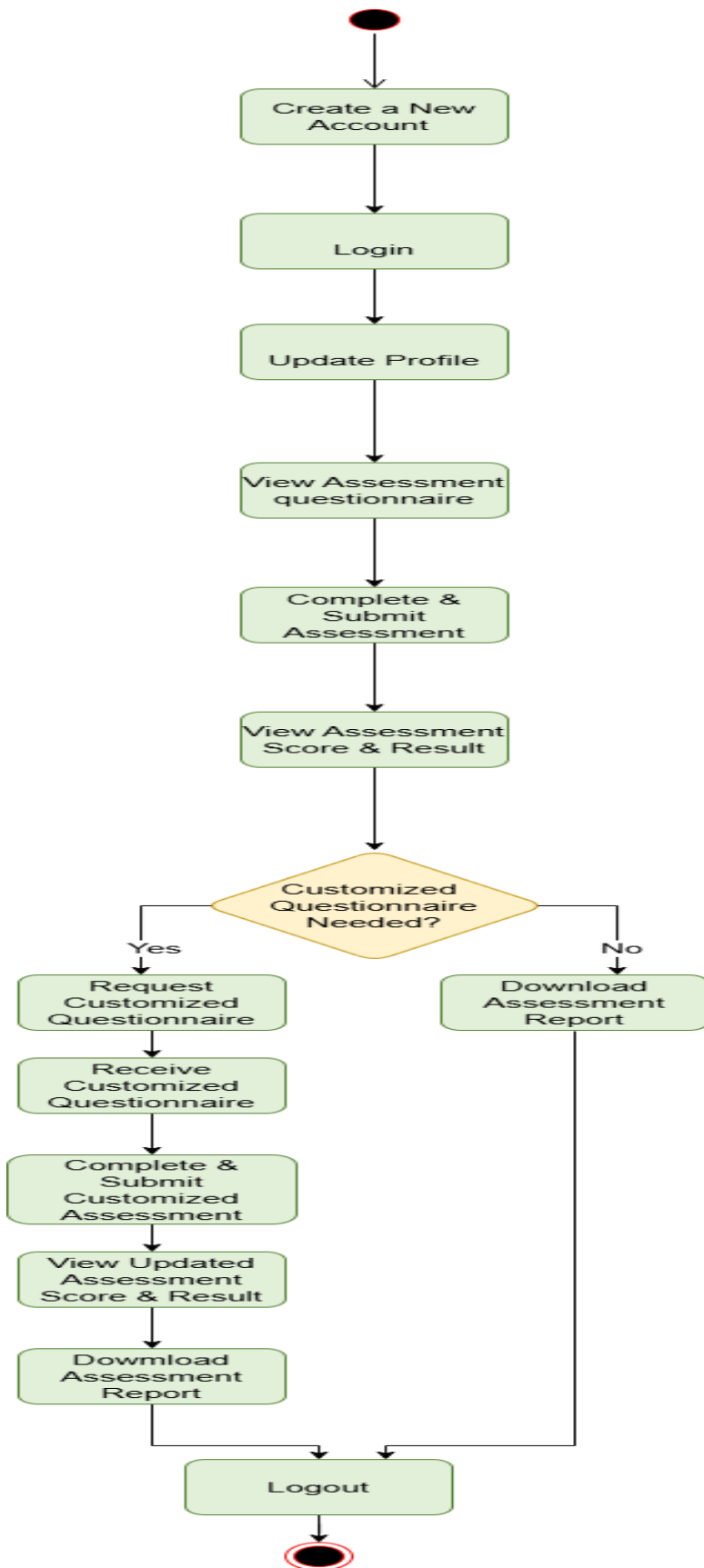


FIG. 1 - CUSTOMER'S ACTIVITY DIAGRAM

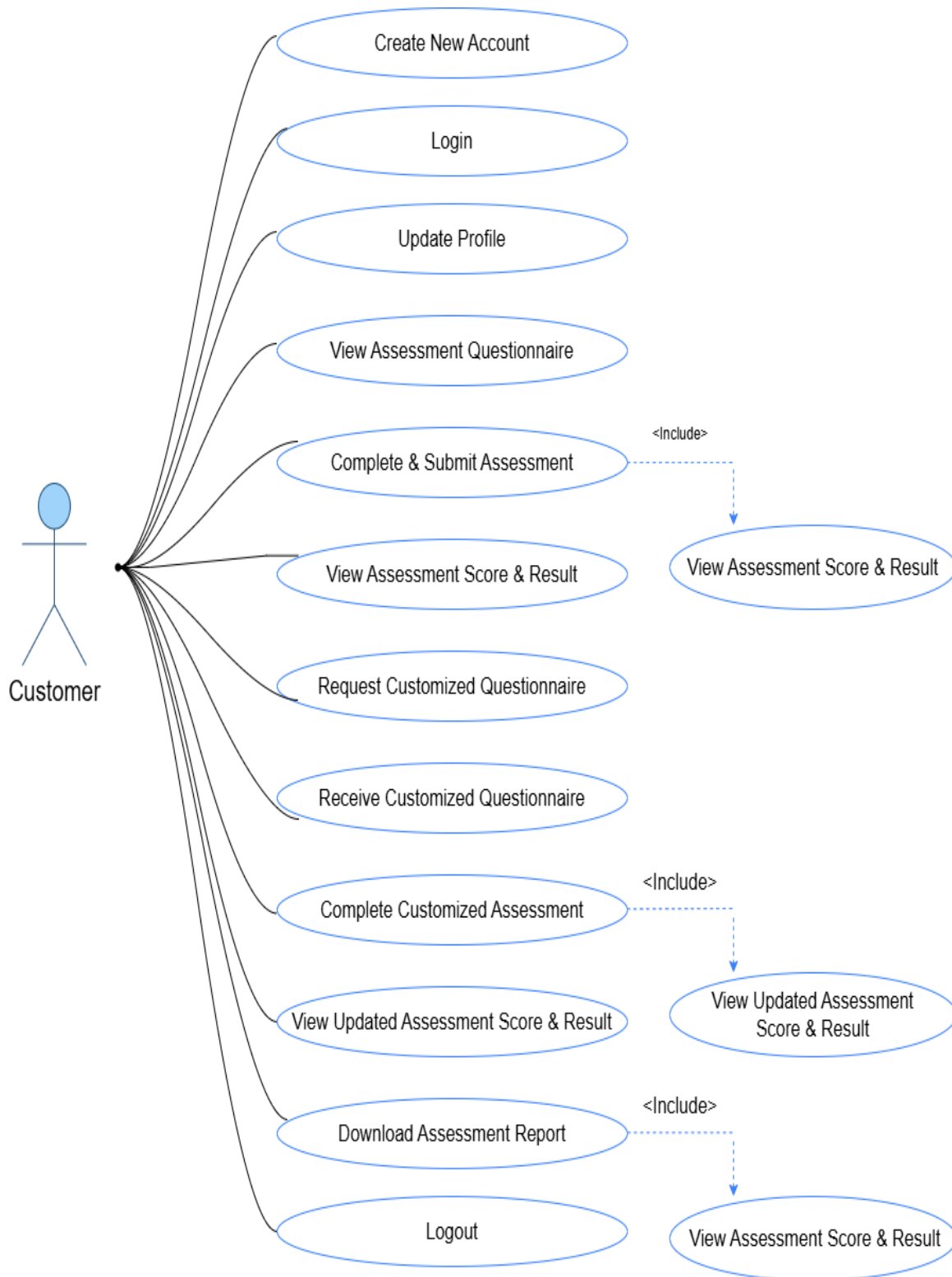


FIG. 1.1 - CUSTOMER'S USE CASE DIAGRAM

2. PSYCHOLOGIST USE CASE:

USE CASE	ACTOR	OBJECTIVE
Register Psychologist Account	Psychologist	Register a new psychologist account in the system.
Login	Psychologist	Access the system securely.
Analyze Assessment Results	Psychologist	Analyze assessment scores, wellness levels, and reports generated by the system.
View Customized Questionnaire Requests	Psychologist	View requests for customized questionnaires from users or organizations.
Prepare Assessment Questionnaire	Psychologist	Prepare a customized assessment questionnaire for a specific mental health activity or target group.
Submit Customized Questionnaire	Psychologist	Submit the customized questionnaire to the customer.
Identify Users Based on Five Assessment Rating Scales	Psychologist	Identify users according to the five assessment rating scales.
View Group Assessment Results	Psychologist	View group assessment results for educational institutions or corporate organizations.
Monitor User Assessment Progress	Psychologist	Monitor users' assessment progress over time.
Provide Assessment Feedback and Wellness Recommendations	Psychologist	Provide assessment feedback and wellness recommendations based on assessment results.
Logout	Psychologist	Exit the system securely.

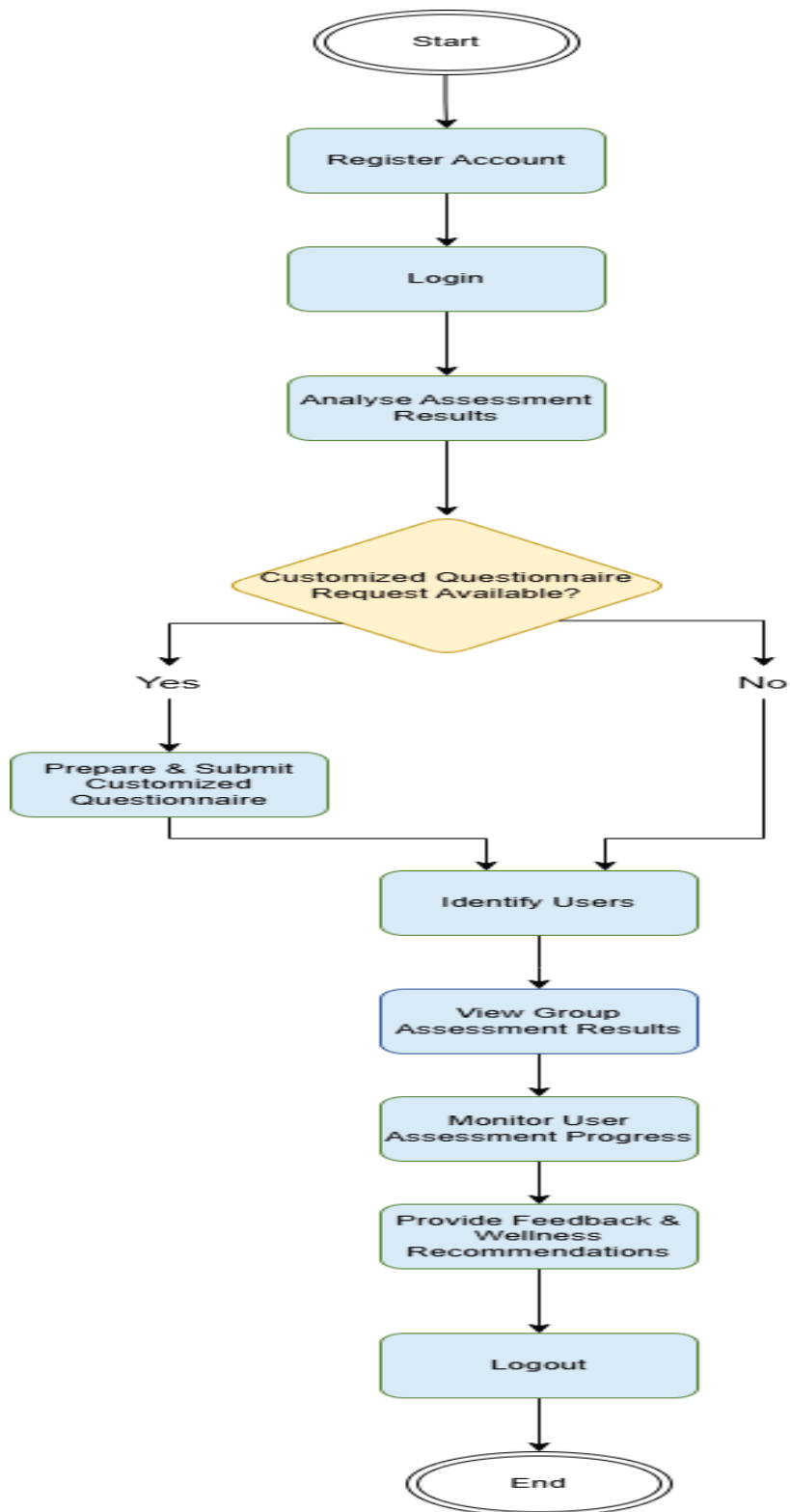


FIG. 2 - PSYCHOLOGIST ACTIVITY DIAGRAM

3. System Administrator Use Case:

USE CASE	ACTOR	OBJECTIVE
Register Administrator Account	System Administrator	Register a new administrator account in the system.
Login	System Administrator	Access the system securely.
View Registered Users	System Administrator	View the list of registered users.
Manage User Accounts	System Administrator	Manage user account information and access.
Manage Assessment Records	System Administrator	Manage and maintain assessment records in the system.
Configure System Settings	System Administrator	Configure and maintain system settings.
Generate Group Assessment Reports	System Administrator	Generate group assessment reports for educational institutions or corporate organizations.
View Previous Assessment Records	System Administrator	View historical assessment records and reports.
Logout	System Administrator	Exit the system securely.

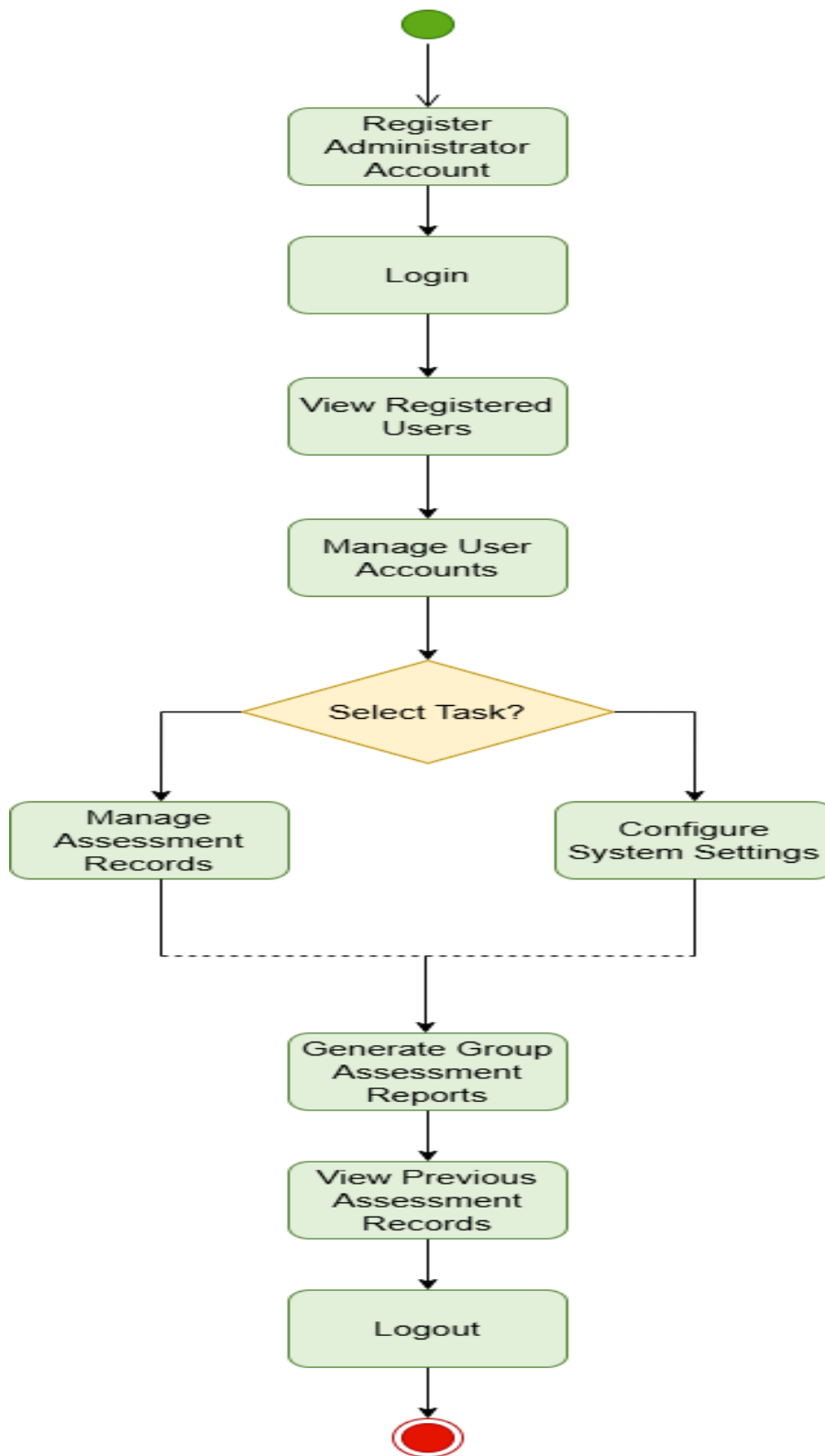


FIG.3 – SYSTEM ADMINISTRATOR ACTIVITY DIAGRAM